B.Tech CSE - Third Year | Lab 9

Assignment 9: Constructors • *Aim: Initialize object state using constructors.*

Experiment No. 9.1: Default Box constructor

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
class Box {
    int width;
    Box() {
        width = 10;
    }
}
public class Task_01_Default_Box_constructor {
    public static void main(String[] args) {
        Box b = new Box();
        System.out.println("Width: " + b.width);
    }
}
```

3. **Output:** `Width: 10`
4. **Viva-Voce:** What is a constructor? When does a constructor run? Does a constructor have a return type?

Experiment No. 9.2: Parameterized Box constructor

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
class Box {
    int width;
    Box(int width) {
        this.width = width;
    }
}
public class Task_02_Parameterized_Box_constructor {
    public static void main(String[] args) {
        Box b = new Box(25);
        System.out.println("Width: " + b.width);
    }
}
```

3. **Output:** `Width: 25`
4. **Viva-Voce:** What is a parameterized constructor? What does this refer to? Why use this.width?

Experiment No. 9.3: Increment a Counter three times

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
class Counter {
    int count;
    Counter() { count = 0; }
    void increment() { count++; }
}
public class Task_03_Increment_a_Counter_three_times {
    public static void main(String[] args) {
        Counter c = new Counter();
        c.increment(); c.increment(); c.increment();
        System.out.println("Count: " + c.count);
    }
}
```

3. **Output:** `Count: 3`
4. **Viva-Voce:** What does count++ do? Why initialize count in the constructor? What is object state?